FUNDAMENTAL THEOREM OF INFINITE GALOIS THEORY: CLOSED SUBGROUPS CORRESPOND TO SUBEXTENSIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The kernel of the restriction map Gal(L/K) -> Gal(E/K) is open because Gal(E/K) is discrete; the finite case E^{Gal(E/K)} = K is theorem-galois-theory of the same chapter.

Editorial note. The label theorem-inifinite-galois-theory names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own theorem environment in the chapter source.

Allowed standard prerequisites (editorial).

- Same chapter theorem-galois-theory (finite fundamental theorem, delivered structure): finite Galois correspondence used throughout without further mention (author's announcement at source lines 3288-3290)
- Same chapter lemma-galois-profinite (the profinite topology on Gal(L/K)), lemma-galois-infinite (canonical surjective continuous restriction map Gal(L/K) -> Gal(E/K)), lemma-galois-goes-up, lemma-normal-closure-inside-normal
- Topology, Lemma topology-lemma-bijective-map is not used here; the closed subgroup argument uses only the topology definition and finite Galois theory

Source reference. Original reference 09I2: sources/stacks-fields/pinned-original-source.tex, source lines 2911–2911. Prerequisite pointer invoked at 3305; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BMK: sources/stacks-fields/pinned-original-source.tex, source lines 3168–3168. Prerequisite pointer invoked at 3296; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BMJ: sources/stacks-fields/pinned-original-source.tex, source lines 3076–3076. Prerequisite pointer invoked at 3274; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BMG: sources/stacks-fields/pinned-original-source.tex, source lines 2351–2351. Prerequisite pointer invoked at 3293; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 09DW: sources/stacks-fields/pinned-original-source.tex, source lines 3001–3001. Prerequisite pointer invoked at 3289; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1 (Fundamental theorem of infinite Galois theory)**.** *Let L/K be a Galois extension. Let $G = Gal(L/K)$ be the Galois group viewed as a profinite topological group (Lemma 0BMJ). Then we have $K = L^G$ and the map*

$$\{\text{closed subgroups of } G\} \longrightarrow \{\text{subextensions } L/M/K\}, \quad H \longmapsto L^H$$

is a bijection whose inverse maps M to $Gal(L/M)$. The finite subextensions M correspond exactly to the open subgroups $H \subset G$. The normal closed subgroups H of G correspond exactly to subextensions M Galois over K.

Proof. We will use the result of finite Galois theory (Theorem 09DW) without further mention. Let $S \subset L$ be a finite subset. There exists a tower $L/E/K$ such that $K(S) \subset E$ and such that E/K is finite Galois, see Lemma 0BMG. In other words, we see that L/K is the union of its finite Galois subextensions. For such an E, by Lemma 0BMK the map $\mathrm{Gal}(L/K) \to \mathrm{Gal}(E/K)$ is surjective and continuous, i.e., the kernel is open because the topology on $\mathrm{Gal}(E/K)$ is discrete. In particular we see that no element of $L \setminus K$ is fixed by $\mathrm{Gal}(L/K)$ as $E^{\mathrm{Gal}(E/K)} = K$. This proves that $L^G = K$.

By Lemma 09I2 given a subextension $L/M/K$ the extension L/M is Galois. It is immediate from the definition of the topology on G that the subgroup $\mathrm{Gal}(L/M)$ is closed. By the above applied to L/M we see that $L^{\mathrm{Gal}(L/M)} = M$.

Conversely, let $H \subset G$ be a closed subgroup. We claim that $H = \mathrm{Gal}(L/L^H)$. The inclusion $H \subset \mathrm{Gal}(L/L^H)$ is clear. Suppose that $g \in \mathrm{Gal}(L/L^H)$. Let $S \subset L$ be a finite subset. We will show that the open neighbourhood $U_S(g) = \{g' \in G \mid g'(s) = g(s)\}$ of g meets H. This implies that $g \in H$ because H is closed. Let $L/E/K$ be a finite Galois subextension containing $K(S)$ as in the first paragraph of the proof and consider the homomorphism $c : \mathrm{Gal}(L/K) \to \mathrm{Gal}(E/K)$. Then $L^H \cap E = E^{c(H)}$. Since g fixes L^H it fixes $E^{c(H)}$ and hence $c(g) \in c(H)$ by finite Galois theory. Pick $h \in H$ with $c(h) = c(g)$. Then $h \in U_S(g)$ as desired.

At this point we have established the correspondence between closed subgroups and subextensions.

Assume $H \subset G$ is open. Arguing as above we find that H contains $\mathrm{Gal}(L/E)$ for some large enough finite Galois subextension E and we find that L^H is contained in E whence finite over K. Conversely, if M is a finite subextension, then M is generated by a finite subset S and the corresponding subgroup is the open subset $U_S(e)$ where $e \in G$ is the neutral element.

Assume that $K \subset M \subset L$ with M/K Galois. By Lemma 0BMK there is a surjective continuous homomorphism of Galois groups $\mathrm{Gal}(L/K) \to \mathrm{Gal}(M/K)$ whose kernel is $\mathrm{Gal}(L/M)$. Thus $\mathrm{Gal}(L/M)$ is a normal closed subgroup.

Finally, assume $N \subset G$ is normal and closed. For any $L/E/K$ as in the first paragraph of the proof, the image $c(N) \subset \mathrm{Gal}(E/K)$ is a normal subgroup. Hence $L^N = \bigcup E^{c(N)}$ is a union of Galois extensions of K (by finite Galois theory) whence Galois over K. □